

Genetic code & Wobble hypothesis

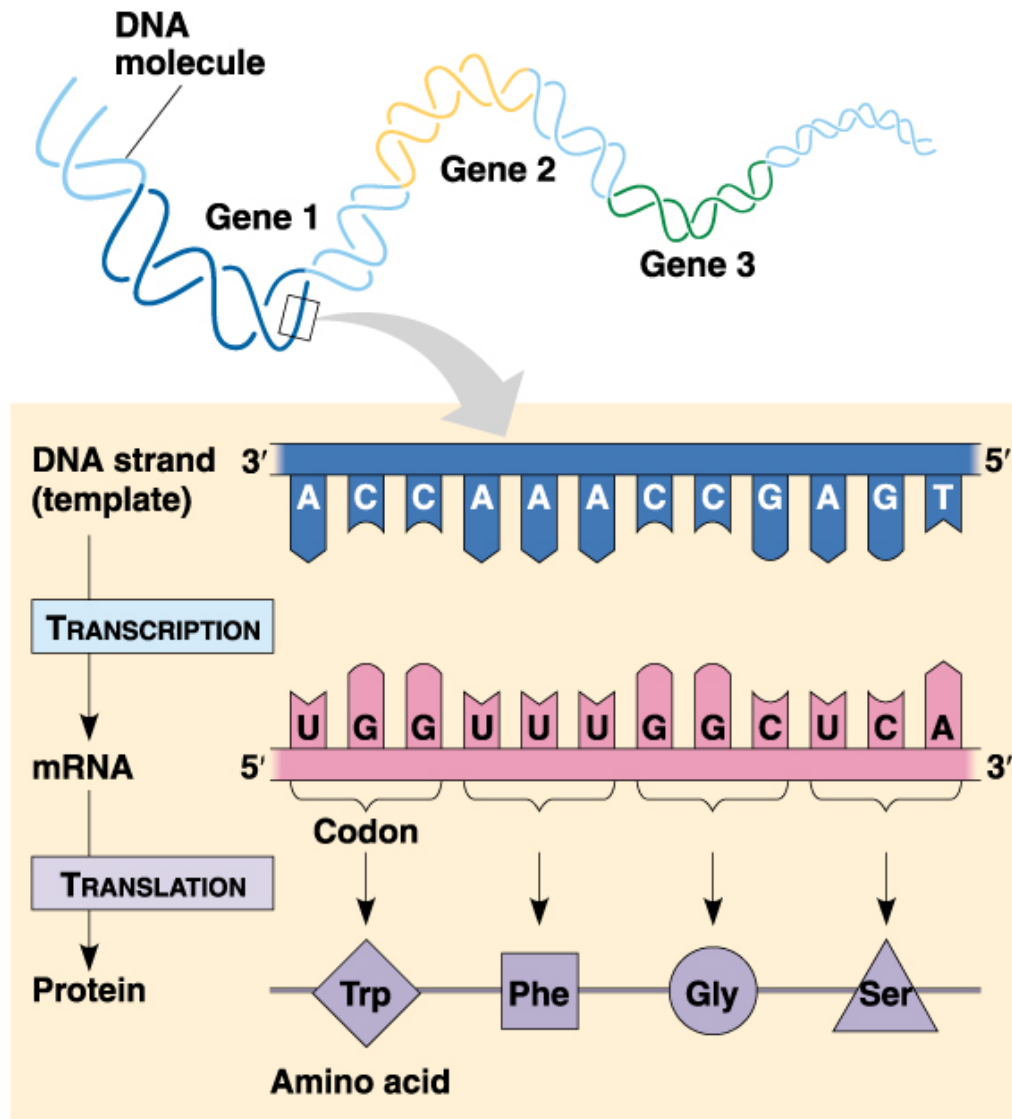
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Objectives:

At the end of this chapter, it is expected;

- To appreciate that the genetic code is a three-letter nucleotide code, which is contained within the linear array of the exon DNA (composed of triplets of A, G, C, and T) of protein coding genes.
- To appreciate that this three-letter code is translated into mRNA (composed of triplets of A, G, C, and U) to specify the linear order of amino acid addition during protein synthesis via the process of translation.
- To appreciate that the universal genetic code is degenerate, unambiguous, nonoverlapping, and punctuation free.
- To describe that the genetic code is composed of 64 codons, 61 of which encode amino acids while 3 induce the termination of protein synthesis.
- To explain the Woobles hypothesis to appreciate single tRNA can recognize more than one codon.

Gene expression



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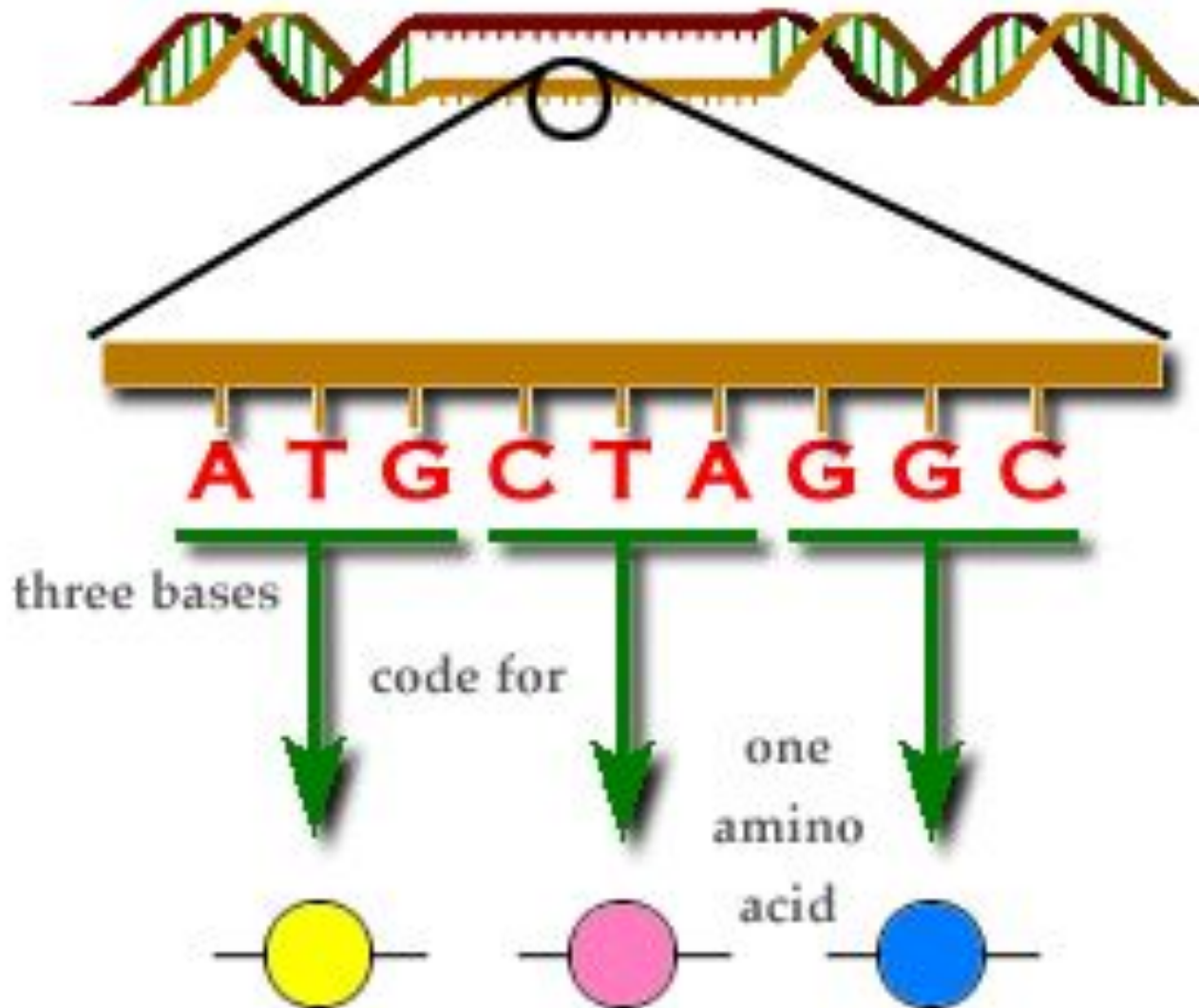
GENETIC CODE

- The genetic code - the set of rules - encoded (information) in genetic material (DNA or RNA sequences) → translated into proteins (amino acid sequences)
- Composed of four nucleotide bases i.e., the Purines – Adenine (A) & Guanine (G), and the pyrimidines – Cytosine (C) & Uracil (C)
- Tri-nucleotide sequences in mRNA called 'Codons' - determine the sequence of amino acids in the proteins
- Out of 64 codons, 61 codons code for 20 amino acids found in protein
- Written from 5' to 3' direction in mRNA.

Discovery:

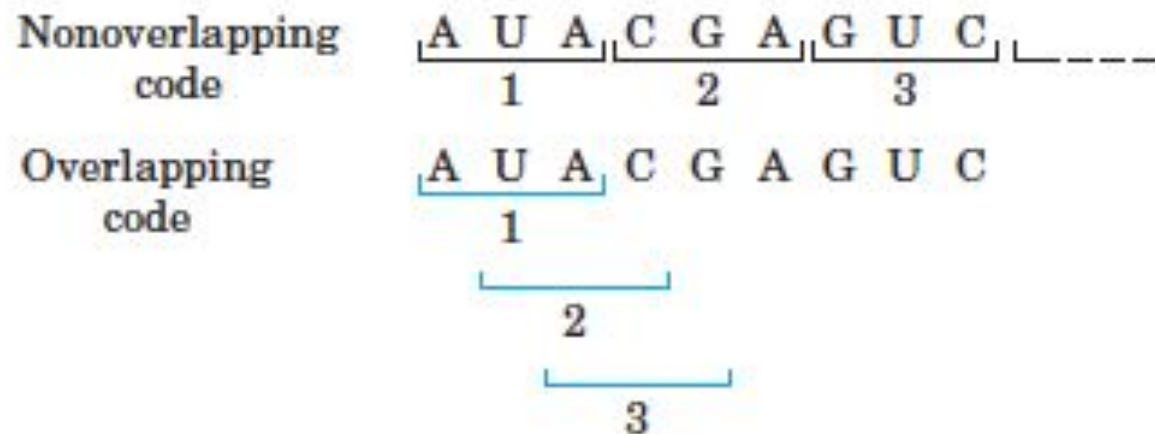
- Philip Leder and Marshall Nirenberg were able to determine the sequences of 54 out of 64 codons
- Har Gobind Khorana identified the rest of the genetic code

The Genetic Code



Characteristics of genetic code:

1. **Universal - almost**, same codon used to code for same amino acids in all the living organisms (some exception, AUA - code for met in mitochondria, but isoleucine in cytoplasm)
2. **Specific** - A particular codon always codes for same aminoacids - highly specific (Unambiguous), e.g. UGG - code for tryp.
3. **Non-overlapping** - read from a fixed point, non-overlapping, commaless, and without punctuations. E.g. UUUCUUAGAGGG is read as UUU/CUU/AGA/GGG. Addition or deletion will radically change the message sequence.



Characteristics of genetic code:

4. **Degenerate** – most of the amino acids have more than one codon b/c 61 codons available to code for only 20 amino acids. E.g. glycine has four codons.

Wobble hypothesis explains codon degeneracy.

Degeneracy of the Genetic Code			
Amino acid	Number of codons	Amino acid	Number of codons
Met	1	Tyr	2
Trp	1	Ile	3
Asn	2	Ala	4
Asp	2	Gly	4
Cys	2	Pro	4
Gln	2	Thr	4
Glu	2	Val	4
His	2	Arg	6
Lys	2	Leu	6
Phe	2	Ser	6

5. **Not punctuated** – there is no punctuation between codons, and the message is read in a continuing sequence of nucleotide triplets until a translation stop codon is reached.

Start/stop codons:

- **Start/stop codons:**
 - Three codons UAA, UAG & UGA - do not code for aminoacids, act as stop signal in protein synthesis - known as termination Or stop codons or nonsense codons.
 - UAG - amber codon**
 - UAA - Ochre codon**
 - UGA - opal codon**
 - The Codon AUG and, sometimes GUG - are the chain initiation codons.

mRNA codons for Amino Acids

		Second base				
		U	C	A	G	
First base	U	UUU } Phenyl- UUC } alanine F UUA } Leucine L UUG }	UCU } UCC } Serine UCA } S UCG }	UAU } Tyrosine Y UAC } UAA } Stop codon UAG } Stop codon	UGU } Cysteine C UGC } UGA } Stop codon UGG } Tryptophan W	U C A G
	C	CUU } CUC } Leucine L CUA } CUG }	CCU } CCC } Proline CCA } P CCG }	CAU } Histidine H CAC } CAA } Glutamine CAG } Q	CGU } CGC } Arginine CGA } R CGG }	U C A G
	A	AUU } Isoleucine I AUC } AUA } AUG } Methionine start codon M	ACU } ACC } Threonine ACA } T ACG }	AAU } Asparagine AAC } N AAA } Lysine AAG } K	AGU } Serine S AGC } AGA } Arginine AGG } R	U C A G
	G	GUU } GUC } Valine V GUA } GUG }	GCU } GCC } Alanine GCA } A GCG }	GAU } Aspartic GAC } acid D GAA } Glutamic GAG } acid E	GGU } GGC } Glycine GGA } G GGG }	U C A G

Reading frame:

- A codon is defined by the initial nucleotide from which translation starts
- Sets the frame for a run of uninterrupted triplets, which is known as an "open reading frame" (ORF).
- For example, the string "CUCAGCGUUACCAU.."
 - if read from the first position, the codons CUC, AGC, GUU, ACC, AU..
 - if read from the second position, the codons UCA, GCG, UUA, CCA, U..
 - if read starting from the third position, CAG, CGU, UAC, CAU..
- Every sequence can, thus, be read in its 5' → 3' direction
- With double-stranded DNA, there are six possible reading frames, three in the forward orientation on one strand and three reverse on the opposite strand

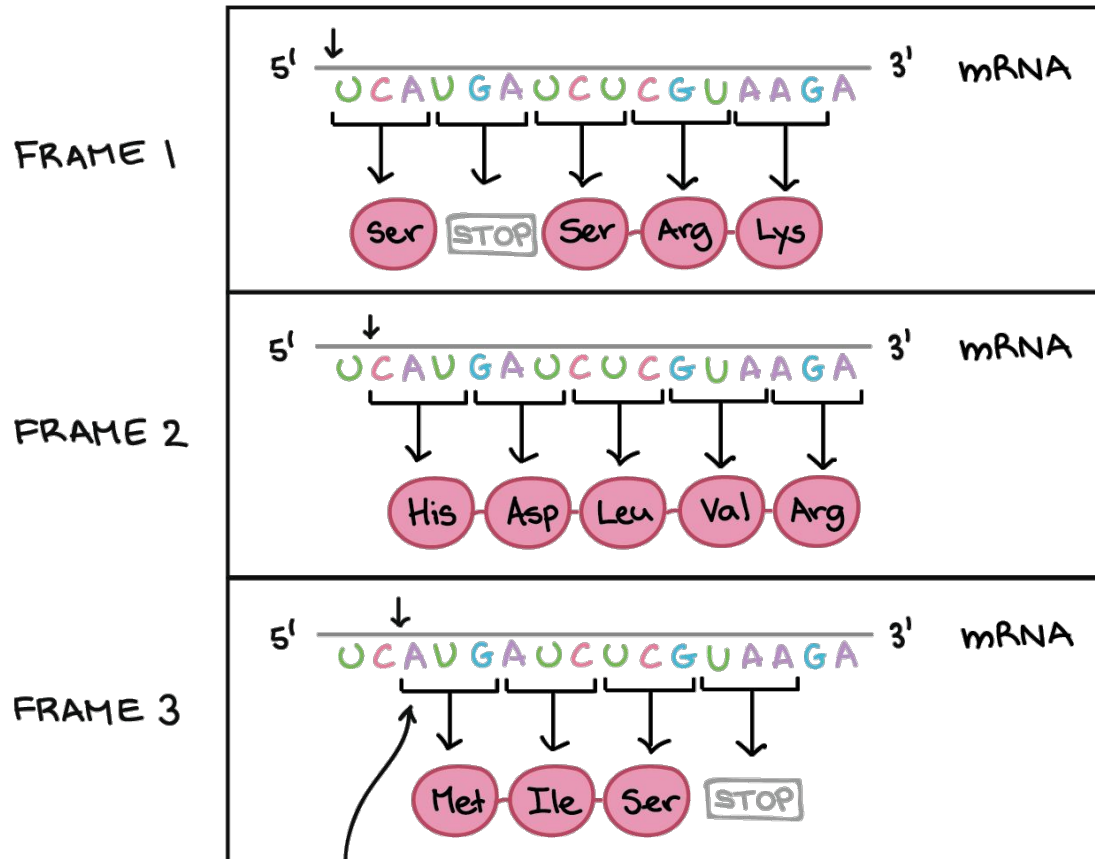
Reading frame:



- One of three possible reading frames contains protein message

Reading frame:

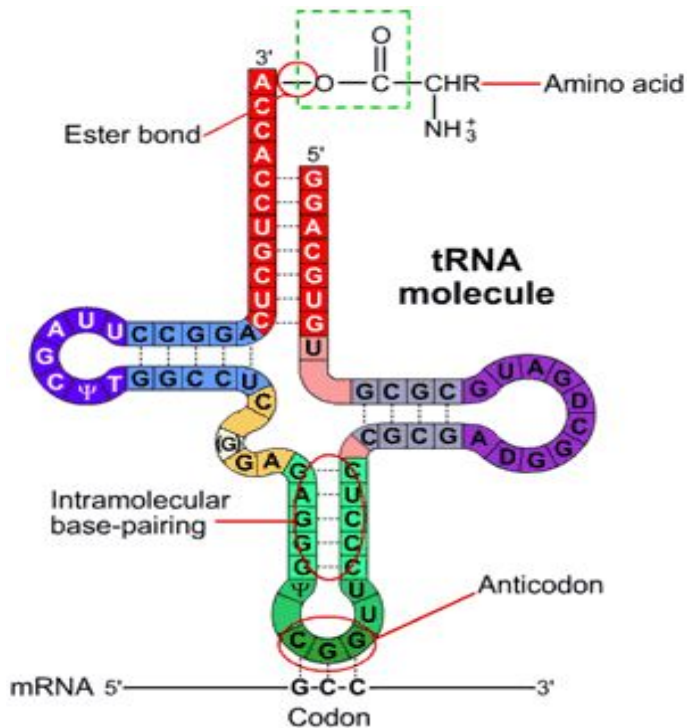
- If the string is "UCAUGAUCUCGUAAGA..";



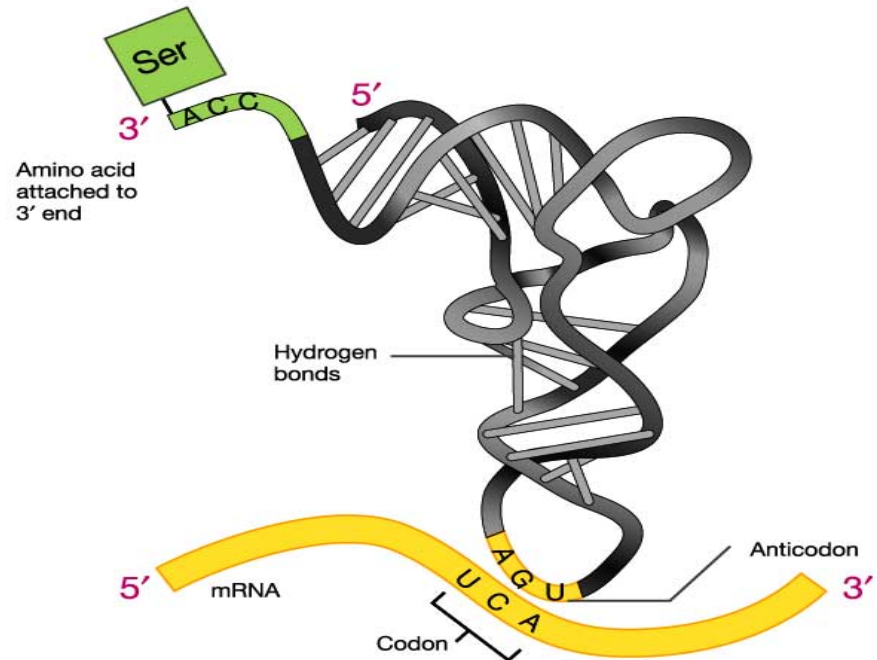
Start codon's position ensures that this frame is chosen

Anticodons & tRNA

Anticodon: is a unit made up of three nucleotides in t-RNA that correspond to the three bases of the codon on the mRNA and involve complementary binding with codon.



Revised model incorporating tertiary structure of tRNA



Conventional complementary pairing ($A=U$, $C\equiv G$) occur b/w first two bases of codon and the last two bases of anticodon. The third base of the codon is rather lenient or flexible with regard to the complementary base.

Wobble hypothesis:

The third base in the codon often fails to recognize the specific complementary base in the anticodon, b/c of this flexibility of the third base in the codon, single tRNA can recognize more than one codon.

Anticodon Codon

C - G

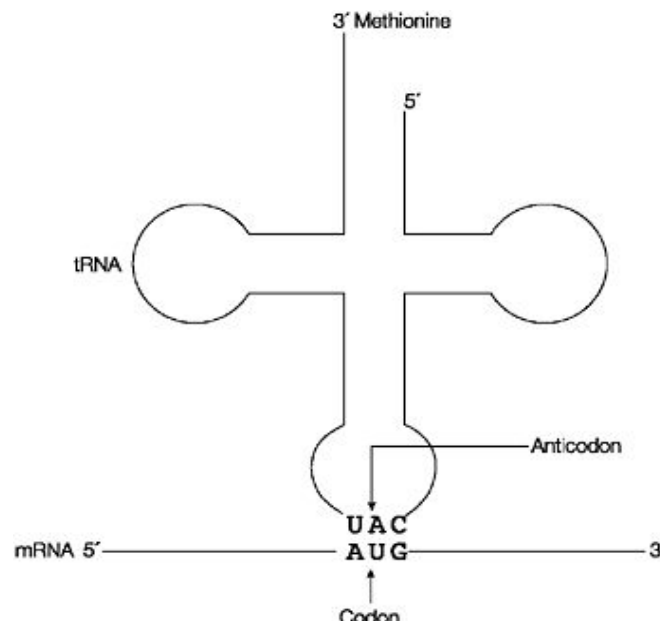
A - U

Conventional base pairing

U - G or A

G - U or C

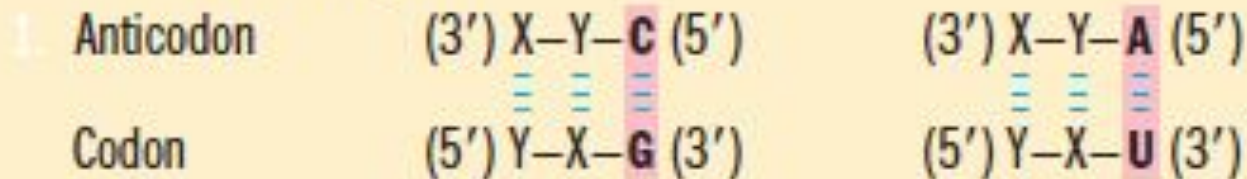
Non-conventional base pairing



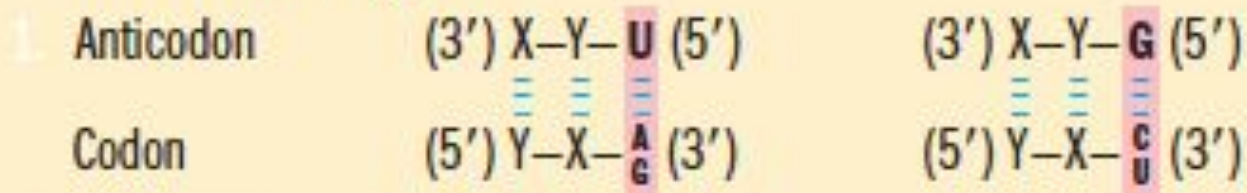
Wobble hypothesis:

How the Wobble Base of the Anticodon Determines the Number of Codons a tRNA Can Recognize

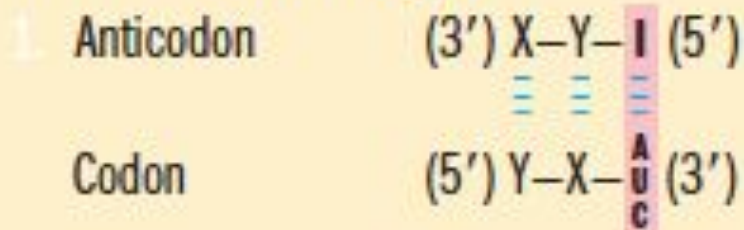
1. One codon recognized:



2. Two codons recognized:



3. Three codons recognized:



Thank you.....



.....Read More!